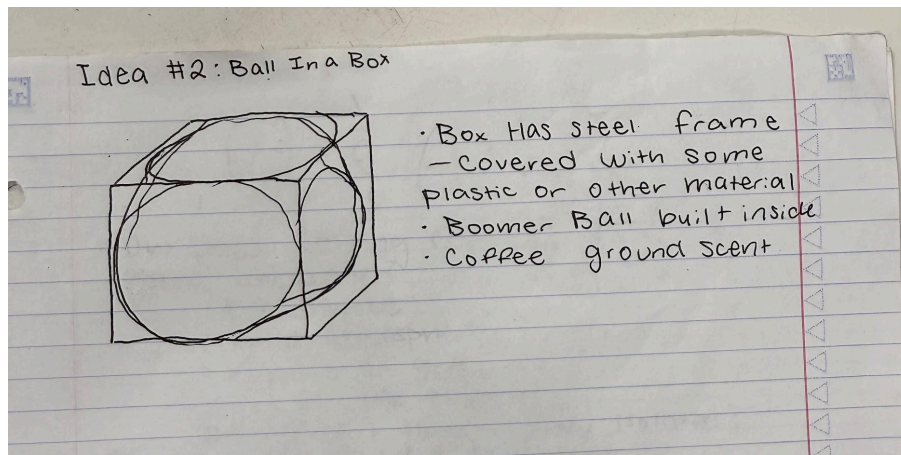
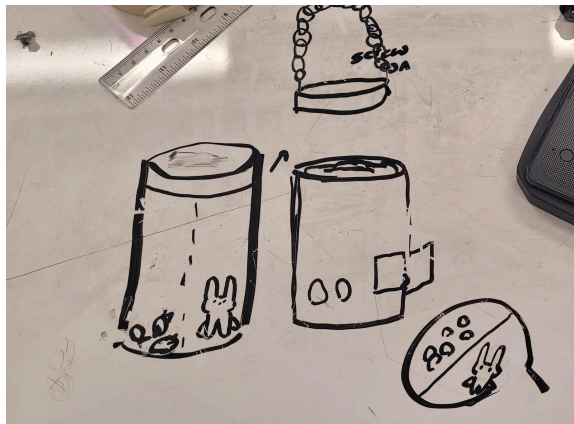


Element D

Solution A is a swinging enrichment toy. The base is made from a large hollowed log. The top being unscrewable for easy access to the inside for the keepers. The log will be divided into two sections through the middle with a plank fitted in which can also be removed for simple cleaning. One section can contain small treats, birds, guinea pigs, or fish. The log will have holes cut into that side for the leopard to swat at and have the pieces fall out. The other side will contain a door that the leopard can swat open to get to a rabbit inside that half. It will be hung by a chain attached by 2 points, which will be connected to the log by 2 screw eyes. This chain will be hung onto a tree branch allowing the leopard to swat at it.



Solution B as shown could also be swatted at by the leopard for a fun bouncing activity. The box exterior would be a steel frame for durability, covered with a softer plastic coating for Natasha's safety. The

Boomer Ball inside would be covered with a scent that we know the Amur Leopards like: Coffee. The idea is that this would entertain them for prolonged periods of time since there is no clear “end” of the activity.



Our third solution would be similar to normal cat toys but upscaled to facilitate a large cat. The concept is a circular hollowed donut kind of shape (shown in image) with a ball fitted inside for the leopard to swat at. The ball will be coffee ground scented to intrigue the leopard to play with it. This replicates a smaller cat toy which has been approved by Larry, the house cat.

		<u>Solution A</u>	<u>Solution A</u>	<u>Solution B</u>	<u>Solution B</u>	<u>Solution C</u>	<u>Solution C</u>
<u>Criteria</u>	<u>Weight</u>	<u>Rating</u>	<u>Weight Score</u>	<u>Rating</u>	<u>Weight Score</u>	<u>Rating</u>	<u>Weight Score</u>
<u>Duration</u>	<u>35%</u>	<u>3</u>	<u>1.05</u>	<u>2</u>	<u>0.7</u>	<u>1</u>	<u>0.35</u>
<u>Durability</u>	<u>15%</u>	<u>2</u>	<u>.3</u>	<u>3</u>	<u>.45</u>	<u>1</u>	<u>0.15</u>
<u>Cleanability</u>	<u>15%</u>	<u>3</u>	<u>.45</u>	<u>1</u>	<u>.15</u>	<u>2</u>	<u>.3</u>
<u>Safety</u>	<u>15%</u>	<u>1</u>	<u>.15</u>	<u>3</u>	<u>.45</u>	<u>2</u>	<u>.3</u>
<u>Weight</u>	<u>10%</u>	<u>2</u>	<u>.2</u>	<u>1</u>	<u>.1</u>	<u>3</u>	<u>.3</u>
<u>Reusability</u>	<u>10%</u>	<u>3</u>	<u>.3</u>	<u>1</u>	<u>.1</u>	<u>2</u>	<u>.2</u>
<u>Overall Score</u>			<u>2.45</u>		<u>1.95</u>		<u>1.6</u>

Our design is a 5 gallon, multi-functional, bucket which hangs on a branch by a chain attached to a rope running through the inside. (Figure 1.) There will be a door cut out of the side with a hinge, attached by stainless steel bolts, nuts, and washers. This door will have a handle also made from our rope as a loop running through 2 holes in the door. This door will also be large enough for a rabbit to fit through if the keeper also wanted to try that. (Figure 2.) On the other side, at the base we will cut out holes large enough for a paw to go into, but not large enough for the head. If you could send me a rough measurement to follow to make sure this dimension is correct that would be greatly appreciated. The main goal for these holes is for Natasha to swat at the bucket and have small animals/pieces of food fall out. The top will be the lid purchased with the bucket, a simple screw on for the keepers to have easy access to the inside.

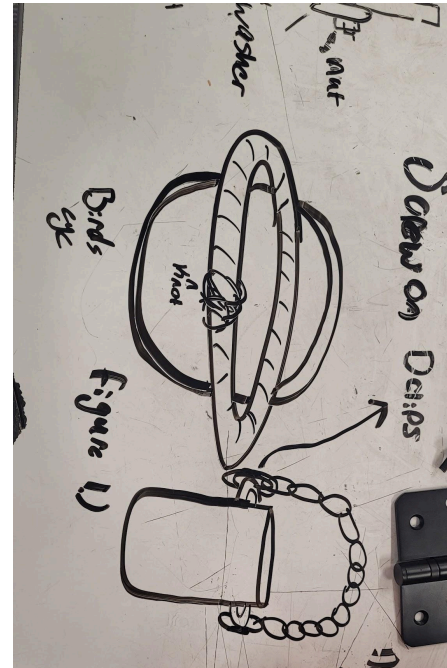
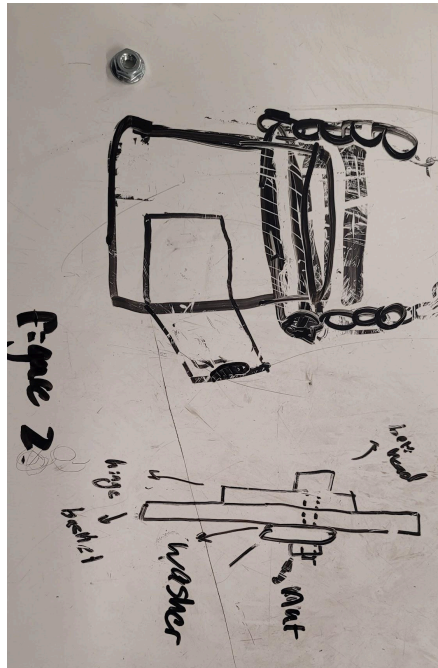
I have attached sketches and prototypes to help understand the concept visually

And also a

CAD

model

Sketch:



Manufacturing Methods:

Step 1: First you will cut out a 8" tall 8" wide rectangle in the middle of the bucket's side

Step 2: Next you will cut 2 ¼ inch holes, 2 inches from the top and bottom of the rectangle, 1 inch from the left side, in the rectangle large enough to fit the rope through

Step 3: Sand all edges that have been cut

Step 4: Cut off a 10 inch piece of the rope, looping it through both holes to create an outside loop, and tie the ends on the inside.

Step 5: Reattach this rectangle using a bolt, hinge, washer, and nut in that order with the bucket in between the hinge and washer, as shown in figure 2.

Step 6: On the other side of the bucket, cut out 2, 5"x4" squares, ½ inch up from the base of the bucket.

Step 7: Sand all edges that have been cut

Step 8: Perpendicular to these holes, cut 2 holes, large enough to fit the rope through on each side of the bucket, each 2 inches apart.

Step 9: Cut 28" of rope, put in these holes to make one large loop with 2 loops on the outside, tie together inside.

Step 10: Attach 2 quick links clips to these loop ropes, and attach our chain,, 13ft, to these links, This can be adjusted to accommodate how tall the keepers want it to hang.

Materials:

Rope: [Rope Link](#)

Washers: [Washers link](#)

Hex Bolt: [Hex Bolt Link](#)

Hex Nut: [Hex Nut Link](#)

Bucket: [Bucket Link](#)

Chain: [Chain](#)

Quick Link: [Quick Link](#)